

41. Which is not true for transition elements

- (a) They are all metals
- (b) They show variable valency
- (c) They form coloured ions
- (d) They do not form co-ordinate compounds**

Answer: (d) They do not form co-ordinate compounds

Explanation: The correct choice is They do not form co-ordinate compounds. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

42. The main reason for not using a mercury electrolytic cell in NaOH manufacture is that

- (a) Hg is toxic
- (b) Hg is a liquid
- (c) Hg has a high vapour pressure
- (d) Hg is a good conductor of electricity**

Answer: (d) Hg is a good conductor of electricity

Explanation: An amalgam is an alloy/solution containing mercury; mercury has distinctive metallic bonding and amalgamation behaviour.

43. Lanthanum is grouped with f-block elements because

- (a) It has partially filled f-orbitals

(b) It is just before Ce in the periodic table

(c) It has both partially filled f and d-orbitals

(d) The properties of Lanthanum are very similar to the elements of 4f block

Answer: (d) The properties of Lanthanum are very similar to the elements of 4f block

Explanation: The correct choice is The properties of Lanthanum are very similar to the elements of 4f block. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

44. The element having electronic configuration belongs to $ns^2(n-1)d^1-10(n-2)f^{1-14}$

- (a) s-block
- (b) p-block
- (c) d-block
- (d) f-block**

Answer: (d) f-block

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

45. Variable valency is shown by

- (a) Typical elements
- (b) Normal elements
- (c) Transition elements**



(d) None of these

Answer: (c) Transition elements

Explanation: Transition elements can use both ns and (n-1)d electrons in bonding because these levels are close in energy, so several oxidation states are possible.

46. Which ion is not coloured

- (a) Cr^{3+}
- (b) Co^{2+}
- (c) Cr^{2+}
- (d) Cu^{+}

Answer: (d) Cu^{+}

Explanation: Partly filled d orbitals allow electronic transitions that absorb visible light; d0 or d10 ions are generally colourless unless charge-transfer effects operate.

47. The number of unpaired electrons in ferrous ion is

- (a) 5
- (b) 4
- (c) 3
- (d) 2

Answer: (b) 4

Explanation: Determine the d-electron count of the atom/ion and apply Hund's rule. More unpaired electrons give a larger spin-only magnetic moment; d0 and d10 species are diamagnetic.

48. Fe, Co and Ni have valuable catalytic properties in process involving

- (a) Organic compound
- (b) Oxidation
- (c) Hydrogenation
- (d) Compounds of hydrogen

Answer: (c) Hydrogenation

Explanation: Transition metals are effective catalysts because they can change oxidation state and form intermediate complexes/adsorb reactants on their surfaces.

49. Which of the following statement is not correct

- (a) Metals contribute their valency electrons to the common sea of electrons
- (b) Metals have high co-ordination number
- (c) Metals tend to adopt closely packed structures
- (d) Metals have high lattice energy

Answer: (a) Metals contribute their valency electrons to the common sea of electrons

Explanation: The correct choice is Metals contribute their valency electrons to the common sea of electrons. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.



50. Zinc, cadmium and mercury show the properties of

- (a) Typical elements
- (b) Normal elements
- (c) Transitional elements**
- (d) Rare elements

Answer: (c) Transitional elements

Explanation: An amalgam is an alloy/solution containing mercury; mercury has distinctive metallic bonding and amalgamation behaviour.

51. Iron is

- (a) A normal element
- (b) A typical element
- (c) A transitional element**
- (d) An inert element

Answer: (c) A transitional element

Explanation: The correct choice is A transitional element. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

52. Platinum, palladium, iridium etc., are called noble metals because

- (a) Alfred Noble discovered them
- (b) They are inert towards many common reagents**
- (c) They are shining lustrous and pleasing to look at
- (d) They are found in active state

Answer: (b) They are inert towards many common reagents

Explanation: The correct choice is They are inert towards many common reagents. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

53. Which of the following statement is not true about Mohr's salt

- (a) It decolourises KMnO_4
- (b) It is a primary standard
- (c) It is a double salt
- (d) Oxidation state of iron is +3 in this salt**

Answer: (d) Oxidation state of iron is +3 in this salt

Explanation: The correct choice is Oxidation state of iron is +3 in this salt. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

54. Which one of the following statement is true for transition elements

- (a) They exhibit diamagnetism
- (b) They exhibit inert pair effect
- (c) They do not form alloys
- (d) They show variable oxidation states**

Answer: (d) They show variable oxidation states



Explanation: The correct choice is They show variable oxidation states. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

55. The valence shell electronic configuration of Cr^{2+} ion is

- (a) $4s^1 3d^4$
- (b) $4s^2 3d^2$
- (c) $4s^2 3d^0$
- (d) $3p^6 4s^2$

Answer: (a) $4s^1 3d^4$

Explanation: Use the Aufbau order with the known transition-metal exceptions; when forming cations, ns electrons are removed before (n-1)d electrons.

56. The hardness of Cr is due to and metallic lustre is due to

- (a) Covalent bond, metallic bond
- (b) Covalent bond, hydrogen bond
- (c) Metallic bond, covalent bond
- (d) Metallic bond, hydrogen bond

Answer: (a) Covalent bond, metallic bond

Explanation: The correct choice is Covalent bond, metallic bond. This follows from the standard periodic and electronic properties of d- and f-block elements tested in this question.

57. In the first transition series, the highest B.P. and M.P. is of

- (a) Cr
- (b) V
- (c) Ni
- (d) Fe

Answer: (a) Cr

Explanation: Melting/boiling trends depend largely on the strength of metallic bonding, which is influenced by the number of valence d and s electrons available for bonding.

58. In the following transition elements, the lowest M.P. and B.P. is exhibited by

- (a) Cr
- (b) Hg
- (c) Cu
- (d) Au

Answer: (b) Hg

Explanation: Melting/boiling trends depend largely on the strength of metallic bonding, which is influenced by the number of valence d and s electrons available for bonding.

59. In the following members of transition elements, the lowest ionization energy is of

- (a) Ti
- (b) Sc
- (c) V
- (d) Mn



Answer: (b) Sc

Explanation: Compare the stability of the resulting electronic configurations as well as effective nuclear charge; half-filled and fully filled subshells often produce anomalies.

60. Which of the following has second ionisation potential less than expected

- (a) Cr
- (b) Zn**
- (c) V
- (d) Mo

Answer: (b) Zn

Explanation: Compare the stability of the resulting electronic configurations as well as effective nuclear charge; half-filled and fully filled subshells often produce anomalies.

